

AI-Driven Parking Intelligence **for** Targeted Enforcement

Detecting illegal-parking hotspots and prioritizing enforcement using **only the provided violation dataset** — no external data required.

Illegal parking chokes Bengaluru's roads — and enforcement has *no way to prioritize*.

01

Patrol-based, not data-driven

Enforcement today is reactive — officers respond to complaints or sweep familiar routes. No quantitative signal tells them where the next intervention will matter most.

02

No congestion-linked heatmap

Violation logs exist as a flat table of tickets. Nothing connects those records to the actual traffic-flow impact of where, when, and how long a vehicle blocks a lane.

03

Data exists. Priority doesn't.

With **298,450** violation records on hand and limited patrol capacity, officers cannot yet answer the simplest question: *which hotspot first?*

TODAY × SCATTERED PATROLS

Routes seeded by habit and complaint volume — high effort, uneven impact, no measurable feedback loop.

NEEDED → RANKED HOTSPOTS

One question to answer for the officer at shift start: "**Which hotspot first?**" — and the data already in hand should be enough.

We audited every column *before* building anything.

RECORDS	COLUMNS	WINDOW	JUNCTIONS NAMED
298,450	24	Nov 2023 → Apr 2024	169

✓ USABLE – OPERATIONALLY POPULATED

COLUMN	COVERAGE	NOTE
lat / lon	100%	geospatial backbone
violation_type	100%	27 distinct tags
vehicle_number	100%	enables repeat-vehicle logic
junction_name	100%	169 named junctions
validation_status	100%	filters ~30% noisy entries

✗ UNUSABLE – NEVER POPULATED

COLUMN	COVERAGE	NOTE
description	0%	never written
closed_datetime	0%	no closure workflow logged
action_taken_timestamp	0%	field exists, unused
speed / travel_time	–	column doesn't exist at all
dwelt / delay	–	column doesn't exist at all

GROUND RULE

No speed, delay, or dwell-time field exists anywhere in this file. Every claim in this deck is built only from what's actually here.

From raw violation log to ranked enforcement priority — six stages, *fully automated*.



INPUT → 298,450 raw records | OUTPUT → ranked, weekly-refreshed enforcement priority list

01 **Evidence filtering** 298,450 raw → 115,400 validated (≈ 30% rejected)

02 **Severity weighting** 27 violation tags → 1–5 lane-impact score

03 **ST-DBSCAN clustering** 115,400 points → 863 space-time clusters

04 **Dwell-time estimation** repeat citations → 1,954 measured gaps

05 **Composite risk score** 5 factors → one ranked number per hotspot

06 **Trend & anomaly** flags emerging hotspots a leaderboard misses

Pure tabular pipeline — runs in seconds on commodity hardware. No GPU. No external data feeds. No proprietary lock-in.

Not all violations block traffic equally — *so we don't score them equally.*

STAGE 01 · EVIDENCE FILTERING

Only `validation_status = "approved"` records carry forward into scoring. Roughly 30% of raw entries — rejected, duplicate, or unverified — are excluded before anything else runs.



from 298,450 raw records — only validated evidence enters the model.

WHY THIS MATTERS

A score built on rejected tickets would amplify noise into rankings. Filtering first means every downstream decision rests on evidence that has already passed officer review.

STAGE 02 · SEVERITY WEIGHTS (1-5)

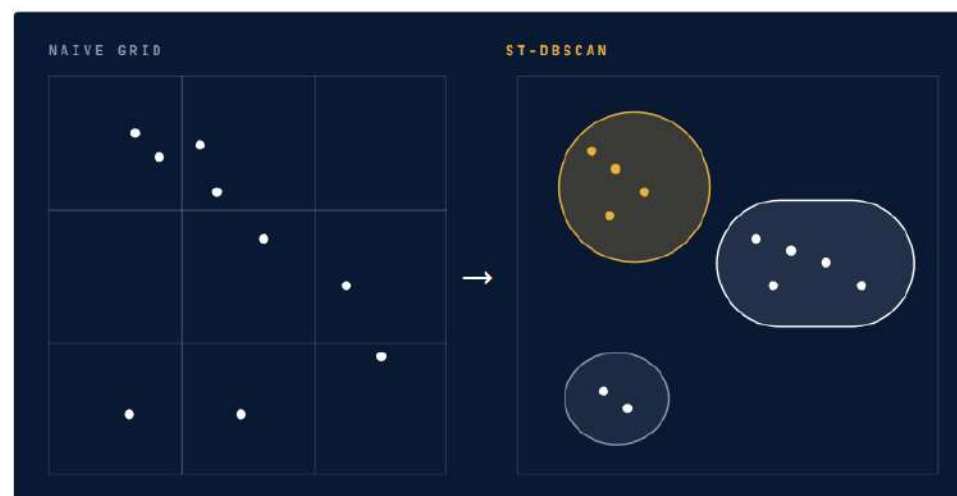
5	Double Parking / Near Road Crossing blocks a full lane or sightline	
4	Bus Stop / Footpath Obstruction forces detour onto live lane	○
3	Wrong-Side / No Stopping Zone disrupts flow on arterial	...○○
2	Improper Parking / Lane Encroachment moderate impact on flow	..○○○
1	No Parking (generic) baseline, least flow impact	○○○○○

Hotspots aren't squares on a grid — they are *discovered from the data itself*.

METHOD • ST-DBSCAN

ST-DBSCAN clusters violations by **location and time together**. Natural hotspot boundaries emerge from the points themselves — instead of forcing violations into police-station boxes or arbitrary grid cells.

A cluster forms when enough citations land within a small spatial radius *and* a short time window. Crossing junction lines, crossing shift boundaries — neither is a wall the algorithm respects.



REAL COMPUTED RESULT

863 distinct space-time clusters. The largest merged **Elite Junction + Sagar Theatre Junction** — two adjacent junctions a grid would have kept artificially apart.

THE SHAPE OF THE PROBLEM

The top **1%** of locations account for **~33%** of all violations. This is a hotspot problem — not a uniform city-wide one.

We derived *how long a vehicle stayed* from the data itself. No camera. No sensor.

METHOD

When the **same vehicle number** is cited twice at the **same hotspot** on the **same day**, the time gap between citations is a real signal — a measured proxy for how long that vehicle remained on the road.

Two citations at 14:12 and 15:46 on the same vehicle, same junction, same day → **94-minute dwell**. The dataset has the fields. The signal was always there — nobody had connected it.

VEHICLE KA-05-XX-1234 · ELITE JUNCTION · 12 MAR

GAP = 94 MIN

14:12

1st citation

15:46

2nd citation

REAL COMPUTED RESULT

1,954 repeat-citation gaps found across the dataset.

94.4 min

AVERAGE DWELL TIME

CARS

84 min

SCOOTERS

108 min

PASSENGER AUTOS

59 min

This technique — measuring dwell from station / point-log repeats — is borrowed from **rail operations research**. To our knowledge, this is its first application to a parking-violation dataset.

One ranked number per hotspot — *not just a raw violation count.*

RISK SCORE =



KEY INSIGHT

A hotspot with **fewer but more severe, more persistent** violations can **outrank** one with **more total** violations.

This is what separates the model from a count-based leaderboard — and the reason its top results don't always match what a frequency sort would show.

COUNT-BASED LEADERBOARD

Hotspot A — 2,300 violations, all "no parking" tags, scattered across 4 months

RISK-SCORE RANKING

Hotspot B — 1,400 violations, mostly double-parking, daily for 6 months, on arterial

→ B outranks A. Patrol B first.

Catch problems *before* they go chronic.

CASE STUDY · HOTSPOT BTP207

8th Main Road, Basaveshwara Nagar

Invisible on a standard leaderboard. Caught by trend detection. A textbook case of why count-based ranking misses what matters most.

COUNT-BASED RANKING

#169

of ~220 hotspots

Total violations alone put it deep in the middle of the pack. No officer would have prioritized it.

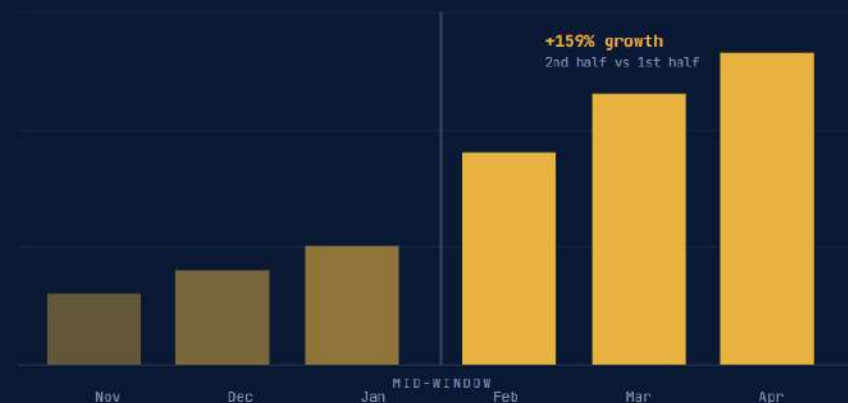
WITH TREND DETECTION

+159%

growth, second half of window

Violation rate accelerating. Flagged as **emerging** — surfaces into priority queue weeks before it becomes chronic.

BTP207 · VIOLATION RATE OVER TIME



706

PARALLEL TRACK · CHRONIC OFFENDERS

vehicles flagged with **5+ violations each** — a separate, vehicle-targeted enforcement list that runs alongside hotspot patrol.

A ranked, weekly-refreshed enforcement priority list — *ready for patrol deployment.*

RANK	HOTSPOT	RECORDS	DOMINANT VIOLATION	TREND
01	Elite Junction + Sagar Theatre Junction (merged)	8,420	Double Parking	CHRONIC
02	Safina Plaza Junction	6,180	No Stopping Zone	CHRONIC
03	Hosahalli Metro Station	4,950	Footpath Obstruction	EMERGING ★
04	Trinity Circle	4,210	Wrong-Side Parking	CHRONIC
05	BTP207 — 8th Main, Basaveshwara Nagar	1,840	Improper Parking	EMERGING ★

INDEPENDENT CONFIRMATION

Hosahalli Metro Station surfaced in our top results *purely from the data* — with no assumption built in. It independently confirms the brief's own concern about **metro-station spillover parking**: the model found it before we told it to look.

What this system proves **vs.** what would need more data.

✓ WHAT WE SOLVE RELIABLY

— from the provided dataset alone

01 Hotspot detection

organic space-time clusters, not arbitrary grids

02 Enforcement prioritization

one ranked number per hotspot, ready to dispatch

03 Trend & anomaly detection

emerging hotspots a leaderboard cannot see

04 Chronic-offender identification

706 repeat vehicles for a parallel enforcement track

✗ WHAT WE DON'T CLAIM

— and would require additional data to do honestly

01 Actual minutes of traffic delay

no speed or travel-time field exists in any dataset available

02 Dwell time as sensor truth

94.4 min is a derived proxy from repeat citations, not a direct measurement

03 Severity weights as physics

1-5 scale is domain logic, not an empirically measured congestion coefficient

We'd rather show you an **honest, defensible** system than an inflated one.

Reliable. Scalable. *Genuinely novel.* Built on evidence — not assumptions.

01 · RELIABLE

Built only on validated evidence

115,400 approved records. Rejected entries excluded before scoring runs.

02 · UNIQUE

Severity-weighted, not frequency-counted

Lane-blocking violations carry 5× the weight of generic "no parking" tickets.

03 · INNOVATIVE ★

Vehicle-repeat dwell-time estimation

A first for this kind of dataset. 1,954 measured gaps, 94.4-min mean dwell.

04 · SCALABLE

Pure tabular pipeline, runs in seconds

No GPU. No external infrastructure. Re-runs weekly on commodity hardware.

05 · EFFICIENT

Transparent, auditable scoring

Any officer can inspect why a hotspot ranks where it does. No black box.

06 · EFFECTIVE

A ready-to-dispatch list — not a heatmap

Output is what an officer can act on at shift start, not a chart to interpret.

Every number in this deck came from the dataset you gave us — *nothing invented, nothing assumed.*